

- Arrays are contiguous blocks of data
- items are only accessible through its indexes

10	13	17	2	4
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↓
 $I(0)$

↓
 $I(l-1)$

* Read: $O(1)$

* Write: $O(1)$

* lookup: $O(N)$

↓, you need to traverse it to find an UnkNwn value

→ Static arrays have fixed size, so when it reaches the declared final size, it can no longer receive any new values without proper handling

↓, handling it though requires creating a bigger / empty array and transfer the old array into the new shell.

→ Writing to an S array in a given index is a $O(1)$ operation, but if you need insert. to happen in a specific order then you might need to shift existing positions to only then be able to write to it.

↳ That would culminate in you having loop over the whole array when shifting, which ends up being a $O(N)$ operation.

→ Same goes here if you want to remove a value/element and still keep you array ordered (No nulls between existing elements)

↳ instead of shifting left for adding a new element to the most right index, we would have to remove the given element (from the random position) and then shift all upcom. elements to the right side.